Muhammad Hatib Umar

HatibUmar@gmail.com
8474090712
www.linkedin.com/in/hatib
Muhammad.Hatib.Umar

SUMMARY: A recent Mechanical Engineering graduate from UIC, with experience and exposure to the manufacturing and design environment. Pursuing growth in roles coupling mechanical design and systems programming.

SKILLS:

Design Principles	3D CAD	Analysis	Programming
Design to Assembly (DTA)	SolidWorks	ANSYS (CFD)	C++
Design to Manufacturing (DFM)	AutoCad	SolidWorks (FEA)	MatLab
Design to Cost (DTC)			HTML

EXPERIENCE:

Zeller Plastik

Manufacturing Operations Intern — ***May 2015 – August 2015***

- Exposed to 5S and Toyota Production Systems Methodologies
- Piloted and proposed various systems to take Plant to greater functionality & efficiency
- Immersed in the world of plastics manufacturing & injection molding
- Demonstrated flexibility and ingenuity in response to fast changing constraints
- Maintained prompt and professional communication

Engineering Design Team (EDT) Project

Design Engineer — ***January 2014 – December 2015***

- Collaborated in building a mobile lift station for the team's autonomous robotics
- Designed manufacture-able and feasible concepts using DFM (Design for Mfg.)
- Worked with design software such as AutoCad and SolidWorks
- Worked in a teamwork environment with great negotiation skills

Collaborated in SMART Initiative for Structuring Organization

Team Member — ***May 2014 – September 2014***

Through SMART (Specific, Measurable, Attainable, Relevant, Time-bound) means proposed a revamped structure for UIC MSA, catering to the relevant demographics, from the ground up

- Coordinated project planning and executed Proposals
- Headed Video editing and presentation, via Camtasia

University of Illinois at Chicago (UIC)

Writing Tutor — ***January 2014 – December 2015***

High literary skill in academic writing classes gave way to working as a writing tutor, after which, excellence in the tutor training class led to a staff position at the UIC Writing Center.

- Developed communication skills, written as well as verbal
- Refined impromptu speaking/teaching/explaining skills
- Collaborated efficiently with a variety of people, personalities, and temperaments
- Built critical and cross-disciplinary thinking

EDUCATION:	**University of Illinois at Chicago (UIC)** Bachelor of Science in Mechanical Engineering (GPA: 3.2/4.0*)*	***December 2015***

PROJECTS

Senior Design: FSAE Intake Manifold ***January 2015-Decemeber 2015***

Designed an intake manifold for the 2016 Formula SAE Racecar, for UIC Motorsports. Primarily taking the Project Manager Role.

- Increased horsepower and torque, via engineering models and CFD Analysis.
- Led the Team to accomplish goals and reached project milestones.
- Assisted CFD (Computational Fluid Dynamics) analysis using SolidWorks and ANSYS.

Refrigeration Process & Maximum Efficiency

Thermodynamic Theory put to the test to determine the actual and maximum efficiency of a household refrigerator. Heat transfer and corresponding values derived via a closed system apparatus within the refrigeration system and utilization of the first law of thermodynamics.

Portable Pixel Map (PPM) Image Editor Program using C++

A program using the PPM image format, arrays, and basic C++ programming to render a robust and unique set of image edits and transformations.

COURSE-WORK:

- Heat Transfer
- Fluid Dynamics
- Machine Design
- Thermodynamics
- Mechanical Vibrations
- Statistics & Probability
- Dynamic Systems and Control
- Industrial Energy Management
- Numerical & Experimental Methods
- Mechanisms & Dynamics/Machinery
- Manufacturing Principles and Materials
- Statics, Dynamics, & Strength of Materials
- Electrical and Computer Engineering/Circuit Analysis

HONORS:

- National AP Scholar
- UIC Dean’s List